

Build Your Own Low-Noise 48V Regulated Power Supply

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Here is a low-ripple and low-noise 48V regulated phantom power supply for an audio preamplifier and driver. Such power supplies require well-filtered, low-ripple and low-noise output voltage with current less than 200mA. These are relatively rare and expensive. Presented here is a simple and low-cost solution suitable for many applications including test equipment and audio applications.

Circuit and working

The circuit diagram of the 48V regulated power supply with low ripple and low noise is shown in Fig. 1. It is built around bridge rectifier BR1, voltage regulator TL783 (IC1), three rectifier diodes 1N4004 (D1 through D3) and a few other components.

TL783 is a high-voltage adjustable voltage regulator that can provide voltage from 1.2V to 125V and output current up to around 700mA. Its internal reference voltage is 1.25V, and minimum output

current is 15mA. That is why the value of resistor R7 is 82ohm ($1.25V/82ohm = 15.24mA$).

Maximum power dissipation of TL783 depends on TO-220 package and its heat-sink. Here, an additional heat-sink with thermal resistance lower than $10^{\circ}C/W$ is used. Output voltage of TL783 can be calculated using the approximate relationship:

$$V_{out} = 1.25V \times (1 + R8/R7)$$

Input 230V AC mains is applied to connector CON1.

AC mains input is protected with fuse F1. Mains AC voltage may have significant noise. Input noise filter is built around capacitors C1 and C2, resistors R1 and R2, and inductors L1 and L2.

Transformer X1 is rated as 230V AC primary to 60V, 1A secondary.

Resistors and capacitors R3, R4, C3, R5, C4 and R6 provide some filtering of the noise from AC mains and from the switching of diodes in BR1. Capacitors C5, C6, C7 and C8 filter the rectified voltage. Also, C9 and C10

PARTS LIST

Semiconductors:

- IC1 - TL783 voltage regulator
- BR1 - W10 bridge rectifier
- D1-D3 - 1N4004 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1, R2 - 10-ohm
- R3, R4 - 1-ohm, 2W
- R5, R6 - 2-ohm
- R7 - 82-ohm, 1W
- R8 - 3.3-kilo-ohm, 1W
- R9 - 4.7-kilo-ohm

Capacitors:

- C1-C4, C6, C10 - 100nF ceramic disk
- C5, C7 - 220 μ F, 220V electrolytic
- C8, C12, C14 - 1nF ceramic disk
- C9, C13 - 10 μ F, 63V electrolytic
- C11 - 220 μ F, 100V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin terminal connector
- X1 - 230V AC primary to 60V AC, 1A secondary transformer
- F1 - 200mA fuse
- L1, L2 - 33mH inductor
- L3, L4 - 100mH inductor
- Heat-sink for TL783
- 230V AC power supply

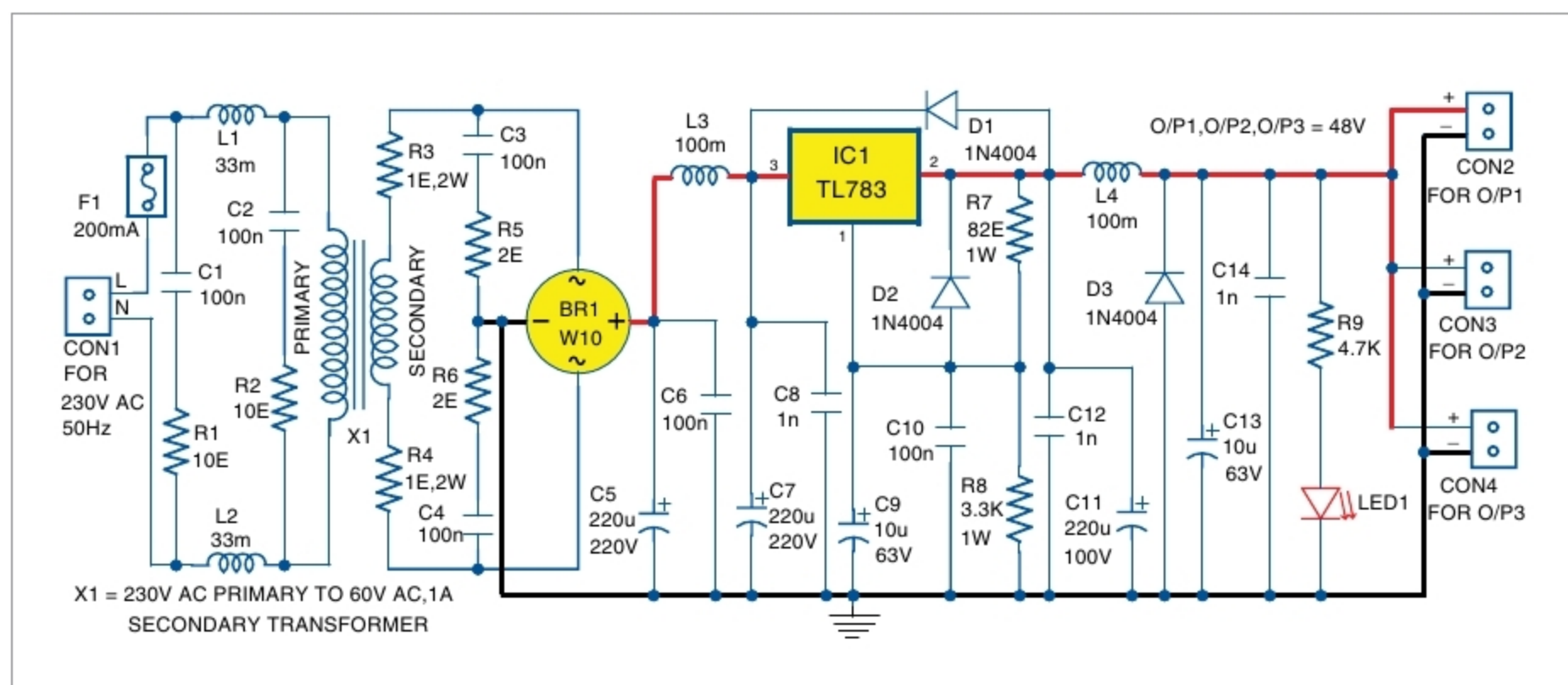


Fig. 1: Circuit diagram of the 48V regulated power supply